

Machine Learning Project Proposal: Speech Emotion Recognition System

Problem Statement

Human communication conveys far more than just words - the emotional tone and delivery of speech carry critical information about a speaker's mental state, intentions, and sincerity.

This project aims to develop an automated Speech Emotion Recognition (SER) system that can classify emotions from audio speech signals. By extracting acoustic features such as Mel-Frequency Cepstral Coefficients (MFCC), pitch, energy, spectral properties, and zero-crossing rates, we can train machine learning models to recognize emotions including anger, happiness, sadness, fear, disgust, boredom, and neutral states.

The primary challenges include: (1) extracting meaningful acoustic features that correlate with emotional states, (2) handling the inherent variability in how different speakers express the same emotion, (3) addressing the imbalanced nature of emotion datasets where some emotions are less frequently expressed, and (4) determining which classification algorithms perform best for this multi-class problem. This project will explore both traditional machine learning approaches (SVM, logistic regression) and compare their performance in emotion classification tasks.

Learning Objectives

Through this project, we hope to achieve the following learning objectives:

- 1. Audio Signal Processing and Feature Engineering:** Develop practical skills in processing raw audio signals and extracting relevant acoustic features.
- 2. Multi-class Classification Implementation:** Apply and compare multiple supervised learning algorithms including Support Vector Machines (with various kernels), Logistic Regression and potentially ensemble methods for classifying seven emotion categories. This includes understanding the theoretical foundations and practical trade-offs of each approach.
- 3. Handling Imbalanced Data:** Develop strategies for dealing with class imbalance commonly found in emotion datasets, where some emotions (like neutral or happy) may be overrepresented while others (like disgust or fear) are underrepresented.
- 4. Dimensionality Reduction and Feature Selection:** Apply Principal Component Analysis (PCA) to reduce the high-dimensional feature space (potentially 30-50 features per audio segment) while retaining discriminative information. Additionally, explore feature selection techniques to identify which acoustic features are most predictive of specific emotions.

5. Model Evaluation with Appropriate Metrics: Develop comprehensive evaluation strategies using cross-validation, confusion matrix analysis, and appropriate metrics (accuracy, precision, recall, F1-score per class, and macro/micro averaging) to assess model performance. Understand why simple accuracy may be misleading with imbalanced classes.

6. Comparative Analysis and Critical Thinking: Compare performance across different algorithms, feature sets, and preprocessing approaches. Develop critical thinking about why certain models perform better for specific emotions and identify systematic errors through confusion matrix analysis.

7. Practical Considerations: Understand real-world constraints including computational efficiency, model interpretability, and the trade-off between model complexity and generalization to unseen speakers.

Reference

<https://www.kaggle.com/code/shoshika/speech-emotion-recognition/notebook>